SOUMYADEEP ROY

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BIO

My work sits at the intersection of biomedical AI, machine learning, natural language processing, and real-world evidence. As medicine continues to generate increasingly complex clinical, genomic, and textual data, this kind of research is central to the future of AI in medicine. I worked with clinical data (structured EHR and unstructured notes) of Parkinson's disease (L3S Research Center, Hannover Medical School, Germany), oncology (breast, lung and prostate cancer) at GE Healthcare India and postoperative pain management at Stanford Medicine.

RESEARCH INTERESTS

Deployable Medical AI · Clinical Decision Support with Real-World Data · Efficient Pretraining and Benchmarking Biological (DNA) Foundation Models · Reliability Evaluation of Clinical LLMs · Guideline-Grounded Medical Reasoning · Auditable AI Systems · Domain Adaptation

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR, Kharagpur, India
PhD, Computer Science and Engineering 07/2025

- Thesis: Domain Adaptation for Medical Language Understanding.

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR, Kharagpur, India
MS, Computer Science and Engineering, CGPA: 9.21 11/2019

- Thesis: Computational Approaches for Online Reputation Monitoring [Modeling, Analysis and Recommendation].

KALYANI GOVERNMENT ENGINEERING COLLEGE, KALYANI, West Bengal, India
B.Tech, Computer Science and Engineering, CGPA: 8.61 06/2017

- Thesis: Understanding Email Interactivity and Predicting User Response to Email.

EXPERIENCE

STANFORD UNIVERSITY, United States
Postdoctoral Scholar, Division of Computational Medicine, Dept. of Medicine 09/2025 - Present

- Faculty Advisor: Prof. Tina Hernandez-Boussard.
- Developing AI models to analyze guideline deviations in real-world patient trajectories
- Improving efficiency, transparency, reliability of reasoning LLMs for high-stakes clinical decision support
- Guest Lecturer, BMDS 223 "Deploying and Evaluating Fair AI in Healthcare" (Spring 2026). Delivered a lecture on "Bias Evaluation in LLMs" (Link to Slides) and another hands-on coding workshop on "Bias Audit on Real-World Data (MIMIC-IV)" (Link to Jupyter Notebook)

WIPRO GE HEALTHCARE PVT. LTD, Bangalore, India
PhD AI Intern, HealthCare Technology & Innovation Center 11/2024 - 05/2025

- Developed generative AI-based dashboard for oncology patient data using Chroma vector database and LangChain, enabling clinicians to efficiently navigate complex patient histories through semantic search.

- Built deep learning-based representation learning system for sensor log data from medical imaging equipment to enable anomaly detection and predictive maintenance. Resulted in one patent filing.
- Collaborated with multi-disciplinary research teams to develop Proof-of-Concept (PoC) models that are ready for engineering teams.

LEIBNIZ UNIVERSITY HANNOVER, Hannover, Germany
Research Associate, L3S Research Center 01/2021 - 06/2023

- Led multi-institutional research collaboration involving doctors and health informaticians, with Hannover Medical School on Parkinson's Disease patient subtyping using Michael J. Fox Foundation LRRK2 dataset (1,500+ patients with clinical (structured EHR) data).
- Developed interpretable decision tree-based patient subtyping method and identified six novel subtypes that were externally validated on the Movement Disorders Society dataset. This work was published in *Frontiers in Artificial Intelligence* journal (Impact Factor: 4.7).
- Developed the pointwise mutual information-based token masking technique for DNA transformer models such as DNABert and LOGO. Achieved 10x-speedup for few-shot gene sequence classification task and got published in 26th European Conference on Artificial Intelligence (ECAI 2023).

ADOBE SYSTEMS INDIA PVT. LTD, Bangalore, India
Research Intern, Big Data Experience Lab 05/2018 – 06/2018

- Built deep learning-based classification model for brand personality detection from website content of Fortune 500 companies as well as a sentence-level text recommendation model to rewrite textual content to adhere to target brand personality. Resulted in one conference (ACM WebSci 2019) and one journal paper (ACM Transactions on the Web, TWEB 2021)
- Designed and executed large-scale data annotation project on Amazon Mechanical Turk with quality control mechanisms.

KEY RESEARCH CONTRIBUTIONS

AI Development and Evaluation with Real World Patient Data

[NeurIPS 2026 Evaluations and Datasets Track: Under Review] Submitted a decision-aware evaluation suite and a novel scale-independent metric, Uncertainty Index, showing that standard uncertainty quantification methods (ensemble and Bayesian) systematically fail to flag high-risk errors in clinical ML across three real-world datasets, exposing critical flaws in uncertainty-based clinical deferral practices.
[Frontiers in AI 2025] Published decision tree-based approach for Parkinson's Disease patient subtyping using clinical and genomic data, validated by medical experts doi code.

Medical LLM Evaluation & Error Analysis

[Under Review] "Cascading Tail Risk: Trajectory-Level Safety Evaluation for Sequential Clinical LLM Agents": Developed "cascading tail risk," a trajectory-level evaluation framework with novel metrics (Tail Harm Escalation and Tail-Restricted Wasserstein Distance) for surfacing rare, compounding safety failures in sequential clinical LLM agents, demonstrated on opioid prescribing in postoperative pain management.
[Under Review] Built Q-Pain++, a factorial stress-testing and multi-modality evaluation framework that audited five clinical LLMs and exposed a decision-space framing effect driving ~92% of treatment decision changes, showing binary safety audits are inadequate for frontier clinical models.
[SIGIR 2024] Conducted systematic evaluation of GPT-4 on USMLE medical licensing exam questions, creating comprehensive error taxonomy identifying 8 distinct failure modes. Released publicly available dataset with 250+ annotated errors doi code.
[ACL 2025 Findings] Demonstrated that high out-of-vocabulary concentration in medical texts leads to significant performance degradation in summarization tasks, with vocabulary adaptation providing effective mitigation doi code.

[WSDM 2025] Co-organized tutorial "Building Trustworthy AI Models for Medicine: From Theory to Applications" covering evaluation protocols, robustness assessment, and deployment considerations doi tutorial website.

[LREC 2026] Created LongTailQA, a disambiguated benchmark addressing 8-30% entity ambiguity in existing long-tail QA datasets, revealing that RAG models gain up to 24.7% accuracy improvement from query disambiguation and demonstrating that retrieval quality matters more than depth for effective long-tail knowledge recall doi code data.

Efficient Model Training & Domain Adaptation

[ECAI 2023] Developed GeneMask, a masked language modeling masking technique, to jointly mask highly correlated tokens, to reduce model pretraining time, and achieve 10x speedup enabling effective few-shot classification of genomic sequences doi code.

[ECAI 2024] Designed adaptive, curriculum masking strategy for DNA transformer models to achieve better performance on genome understanding evaluation dataset, for both full dataset and few-shot gene sequence classification doi code.

[IJCAI 2024] Proposed vocabulary adaptation method for medical text summarization, reducing fine-tuning time from 450 days to 45 hours through fragment score-based hyperparameter optimization doi code.

[EMNLP 2024 Findings] Developed adaptive BPE tokenization technique that modifies initialization step to better preserve domain-specific terminology during fine-tuning doi code.

[ACL 2026 Mains] "Learning Faster with Better Tokens: Parameter-Efficient Vocabulary Adaptation for Specialized Text Summarization." Augments pretrained tokenizers with domain-specific tokens while selectively replacing under-trained, unreachable tokens to limit parameter growth. Reduce training time by 35-55% over continual pretraining and reduce parameter counts by 37% (to appear, San Diego, July 2026).

Clinical NLP Applications

[CIKM 2021] Built knowledge-aware dual-encoder neural network for medical forum question classification, prioritizing medical concept-bearing words extracted using the QuickUMLS tool doi code.

[ICDH 2023] Developed interpretable clinical trial search system using PubMed citation network to improve retrieval by metapath-based similarity search and improved aspect-based ranking fusion (Best Student Paper Candidate) doi code.

[ECAI 2025] Mentored a PhD student through a multi-institution collaboration, creating annotated dataset for physician intent modeling in doctor-patient dialogues and showing intent filtering improves dialogue summarization performance doi code.

APPLIED & TRANSLATIONAL WORK

[Stanford PostDoc Project, 2025 - present] Develop computable guideline pathways and compute deviations in patient care trajectories using Stanford Healthcare and MIMIC data (structured EHR and clinical notes)

[Boston GrandHack 2026] Co-developed AuriCare, a holistic pain management decision support concept, presented at MIT Hacking Medicine's Boston GrandHack 2026 blog event page

[Patent, filed in 2025] Co-inventor on patent filing for deep learning-based representation learning system for medical imaging equipment sensor logs, enabling anomaly detection and predictive maintenance (Wipro GE Healthcare).

TECHNICAL SKILLS

Deep Learning: PyTorch, Transformers, Parameter-Efficient Fine-Tuning (LoRA, QLoRA).
NLP & LLMs: Fine-tuning, Prompt Engineering, RAG Systems, Model Evaluation, Efficient Training.
Healthcare AI: Clinical NLP, EHR Analysis, Medical Knowledge Graphs (UMLS, PrimeKG), MIMIC-III/IV.
ML Tools: Weights & Biases, Docker, Git.
Programming: Python, R, SQL, Bash.
Data Collection & Analysis: Potato, Amazon Mechanical Turk, Prolific.

PROFESSIONAL ACTIVITIES

Memberships

- Sigma Xi, The Scientific Research Honor Society, Association for Computational Linguistics, AMIA Digital Member

Reviewing

- Reviewed for CORE A* CS conferences: EMNLP 2022, ACL 2023, EMNLP 2023, AAAI 2023 to 2026, ACL Rolling Review February, May, July (2025), January, March (2026), FAccT 2026.
- Reviewed for CS CORE A conferences: COLING 2025, ML4H 2024, EACL 2023, ECML-PKDD 2020 (CORE A).
- Reviewed for Medical AI and Informatics conferences and journals – AMIA 2026, Journal of the American Medical Informatics Association, Frontiers in AI, Knowledge and Information Systems, Frontiers in Genetics, IEEE Transactions on Cognitive and Developmental Systems, Elsevier Artificial Intelligence in Medicine.

Tutorials/Invited Talks

- January 2026: December 2024: Delivered an invited talk on "Foundation Models for Medical Text", for the course "AI Foundation Models in Biomedicine" of Leibniz University Hannover, Germany Link to slides
- January 2026: Invited talk titled "Building a Career in AI for Industry and Academia - Current and Future trends" organized by the Google Developers Group on Campus Kalyani Government Engineering College - Kalyani, India Link to website, Link to slides.
- December 2025: Invited talk titled "Vocabulary Adaptation Strategies for Finetuning Medical Language Models - Towards Reliable Medical Summarization" delivered at the Breakfast Talk series of Microsoft Research India Bangalore, India Link to slides.
- August 2025: Invited to present ECAI 2023 and 2024 work on "Improving Training Efficiency of DNA Foundation Models" at the 1st Symposium on Intersections: Statistical Genomics and Complex/Chronic Disease Biology, organized by BRICS - National Institute of Biomedical Genomics, Kalyani, West Bengal.
- March 2025: Conference tutorial titled "Building Trustworthy AI Models for Medicine: From Theory to Applications" delivered at the 18th ACM International Conference on Web Search and Data Mining (WSDM 2025) Tutorial website, Link to slides.
- December 2024: Delivered an invited talk on "Foundation Models for Medical Text", for the course "AI Foundation Models in Biomedicine" of Leibniz University Hannover, Germany Link to slides.

Teaching

- **Stanford University:** Guest Lecturer, BMDS 223 "Deploying and Evaluating Fair AI in Healthcare" (Spring 2026). Link to course website May 5 Lecture, May 7 Coding Workshop
- **IIT Kharagpur:** Programming and Data Structure Lab (Spring 2018), Department Website Team (Autumn 2018 to Spring 2020), Information Retrieval (Autumn 2020), Computing Lab 1 (Autumn 2023).
- **Leibniz University Hannover, Germany:** Natural Language Processing (Winter 2021), Foundations of Information Retrieval (Summer 2022, 2023).